Model Performance Report

RecoMart Recommendation Pipeline | Task 9 & 10 Deliverable

1. Model Tracking Overview

All model training runs are tracked using MLflow. The MLflow tracking server records hyperparameters, evaluation metrics, and the trained model artifacts in a SQLite database (mlflow.db). This enables performance comparison across multiple training iterations.

2. Content-Based Recommendation Model

Algorithm: Gradient Boosting Regressor (Scikit-Learn)

Task: Predict the probability of a product being recommended based on product features.

2.1 Evaluation Metrics

Metric	Description	Value
RMSE	Root Mean Squared Error (Lower is better)	0.07536
MAE	Mean Absolute Error (Lower is better)	0.04429
R2 Score	Coefficient of Determination (Higher is better, Max 1.0)	0.93862

Conclusion: The content-based model performs exceptionally well, explaining ~93.8% of the variance in recommendation probabilities.

3. Collaborative Filtering Model

Algorithm: Singular Value Decomposition (SVD) via Surprise Library

Task: Predict explicit user ratings for products using the user-item interaction matrix.

3.1 Evaluation Metrics

Metric	Description	Value
RMSE	Root Mean Squared Error (Rating scale 1-5)	0.9941
MAE	Mean Absolute Error	0.7906
Precision@10	Fraction of relevant items in top-10 recommendations	0.4629
Recall@10	Fraction of total relevant items retrieved	0.4784
NDCG@10	Ranking quality score	0.6952

Conclusion: The SVD model exhibits strong ranking performance (NDCG@10 = 0.6952) thanks to the dense overlap of users and popular items in the synthetic interactions dataset. The RMSE (0.9941) shows reliable explicit rating predictions as well.

4. Tracked Model Metadata

The following metadata is stored in MLflow for the latest run:

Run Name	Run ID (Short)	Status

collaborative_filtering_model	568205e7	FINISHED
content_based_model	d03ee9ec	FINISHED